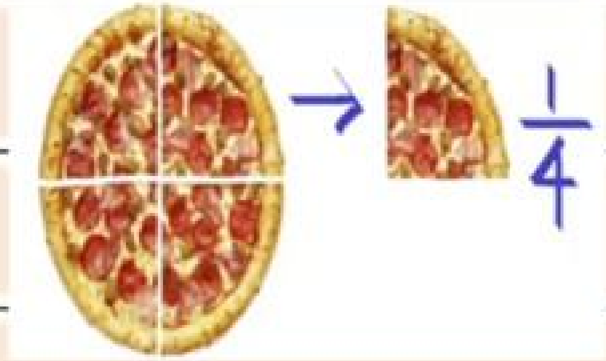
CONCEPT 01

$\frac{1}{2}$ = 50%	$\frac{1}{8}$ = 12.50% or $12\frac{1}{2}$%
$\frac{1}{3}$ = 33.33% or $33\frac{1}{3}$%	$\frac{1}{9}$ = 11.11% or $11\frac{1}{9}$%
$\frac{1}{4}$ = 25%	$\frac{1}{11}$ = 9.09% or $9\frac{1}{11}$%
$\frac{1}{5}$ = 20%	$\frac{1}{12}$ = 8.33% or $8\frac{1}{3}$%
$\frac{1}{6}$ = $16\frac{2}{3}$%	$\frac{1}{14}$ = 7.14% or $7\frac{1}{7}$%
$\frac{1}{7}$ = 14.28% or $14\frac{2}{7}$%	$\frac{1}{15}$ = 6.66% or $6\frac{2}{3}$%



NOTE :

$$\frac{1}{3} = 33\frac{1}{3}\% \Rightarrow \frac{2}{3} = 66\frac{2}{3}\%$$

$$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$$

CONCEPT 01

$\frac{1}{7} = 14.28\%$

$\frac{1}{14} = 7.14\%$

$\frac{1}{2} = 50\%$	$\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$
$\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$	$\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$
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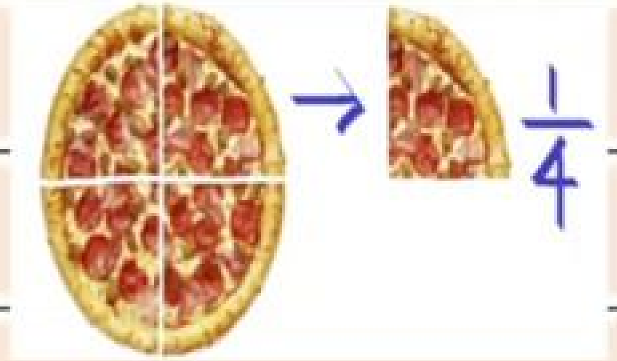
$\frac{1}{6} = 16\frac{2}{3}\%$	$\frac{1}{14} = 7.14\%$ or $7\frac{1}{7}\%$
$\boxed{\frac{1}{7} = 14.28\% \text{ or } 14\frac{2}{7}\%}$	$\frac{1}{15} = 6.66\%$ or $6\frac{2}{3}\%$



NOTE :

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$$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$$

$\frac{1}{9} = 11.11\%$

$\frac{1}{11} = 09.09\%$

CONCEPT 01

$\frac{1}{4\times2} = 12.05$

$\frac{1}{2} = 50\%$	$\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$
$\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$	$\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$
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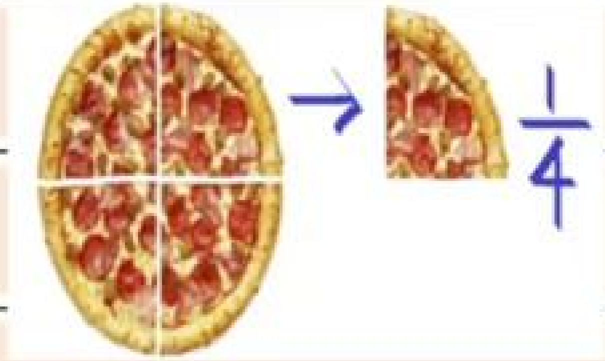
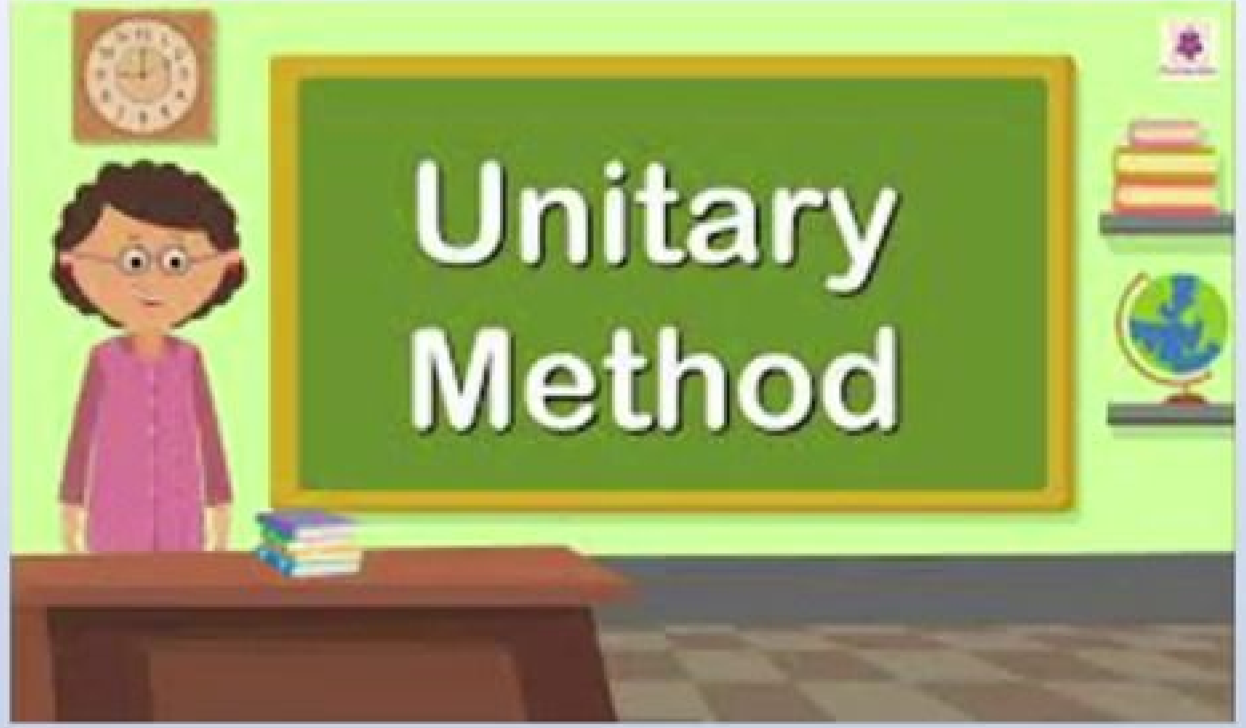


NOTE :

$$\frac{1}{3} = 33\frac{1}{3}\% \Rightarrow \frac{2}{3} = 66\frac{2}{3}\%$$

$$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$$

$\# \; \frac{1}{8} = 12.5 \longleftrightarrow \frac{1}{12} = 8.33\%$



25% of a number is 20.what is 40% of that number also find the number ?

$$\frac{20}{\cancel{25}} \times \overset{4}{\cancel{100}} = 80$$

- If the price of one Kg of wheat is increased by $12\frac{1}{2}\%$, the increase is Rs.5 . Find the original and new price of wheat per Kg ?

$$12\frac{1}{2}\% = +\frac{1}{8}$$

40 Rs

$$I_1 = 8$$
$$I_2 = 9$$

$$1 = 5$$

45 Rs

- A and B appeared in an examination . If the difference of their marks is 25 and percentage of their marks is given as 20 %. Find the full marks for which examination has been held ?

$$\frac{25}{20} \times 100$$

- If the price of one Kg of rice is increased by 25%, the increase is Rs 12. Find the new Price of rice per kg ?

$$25\% = \frac{1}{4}$$

$$I_1 = 4$$
$$I_2 = 5$$

$$1 = 12$$

60 Rs

- A and B appeared in an examination . If the difference of their marks is 25 and percentage of their marks is given as 20 %. Find the full marks for which examination has been held ?

$$\frac{25}{20} \times 100$$

- If the price of one Kg of rice is increased by 25%, the increase is Rs 12. Find the new Price of rice per kg ?

$25\% = \frac{1}{4}$

$I_1 = 4$

$I_2 = 5$

$1 = 12$

60 Rs


Moms Rule!


Dad Rule #11
'Adult Language'
is a privilege reserved for
adults. When you have a
full-time job and a mortgage
you can say any damn thing
you want to say

$\uparrow$ { $20\% \uparrow$ — $\times \frac{120}{100}$

$30\% \uparrow$ — $\times \frac{130}{100}$ }

$\downarrow$ { $10\% \downarrow$ — $\times \frac{90}{100}$

$20\% \downarrow$ — $\times \frac{80}{100}$ }

$20\% \uparrow = \frac{1}{5}$ $\left(\times \frac{6}{5}\right)$

$20\% \downarrow = \frac{1}{5}$ $\left(\times \frac{4}{5}\right)$

$33.33\% \uparrow = \frac{1}{3}$ $\times \frac{4}{3}$

$\downarrow$ $\left(\times \frac{2}{3}\right)$

CONCEPT 02:

1. Percentage Change $= \dfrac{\text{Final} - \text{Initial}}{\text{Initial}} \times 100$

2. Quantity I is how much percent of Quantity II

$$\frac{\text{Quantity I}}{\text{Quantity II}} \times 100$$

3. Quantity I is how much percent more than Quantity II

$$\frac{\text{Quantity I} - \text{Quantity II}}{\text{Quantity II}} \times 100$$

4. Quantity I is how much percent less than Quantity II

$$\frac{\text{Quantity II} - \text{Quantity I}}{\text{Quantity II}} \times 100$$

7

1. What percent is 2 minutes 24 seconds of an hour?

a. 6%
b. 2%
c. 4%
d. 8%

I F

$2 \longrightarrow 3$

% change $= \frac{+1}{2} \times 100 = 50\%$

8Q If Arjun's salary is 20% more than that of Bheem, then how much percent is Bheem's salary less than that of Arjun?

~~1) 16.66%~~ 2) 20% 3) 40% 4) 10% 5) None of these

$$\left[\frac{R}{100 \pm R} \times 100\right]$$

Arjun	Bheem
↓	↓
120	100

$$\frac{-20}{120} \times 100$$

Q. If Arjun's salary is 20% more than that of Bheem, then how much percent is Bheem's salary less than that of Arjun?

1) 16.66% 2) 20% 3) 40% 4) 10% 5) None of these

$$\left[\frac{20}{120} \times 100\right]$$

Arjun → 120

Bheem → 100

$$\frac{-20}{120} \times 100$$

9Q. If A's salary is 25% less than B's salary, then B's salary is what % more than A's salary?

1) Cannot be determined   2) 21   ~~3) 33.33~~   4) 25
5) 20

$\frac{1}{4}$

A ↓   B ↓
3   4

$\frac{+1}{3} \times 100$

10 Q. Journalist Popatlal's income is 37.5% less than Dr. Iyyer's. Then Dr.Iyyer's income is how much % more than Popatlal's income?

$-\frac{3}{8}$

P	I
5	8

$\frac{+3}{\cancel{5}} \times \cancel{100}^{20} = 60\%$

11Q. The population of a town is 18000. It increases by 10% during first year and by 20% during the second year. The population after 2 years will be ?

(A) 19800 (B) 21600 (C) 23760 (D) None

$$18000 \times \frac{110}{100} \times \frac{120}{100} = \_\_\_\_\_\_$$

Q. The population of a town is 18000. It increases by 10% during first year and by 20% during the second year. The population after 2 years will be ?

(A) 19800 (B) 21600 ~~(C) 23760~~ (D) None

9

$180 \times \underline{11} \times \underline{12} = \_\_\_\_$

0 0

21 6 0 1

760

12. **The value of lathe machine depreciates at the rate of 10 % per annum. If the cost of machine at present is Rs. 160,000, then what will be its worth after 2 years?**

a. Rs. 122,365 b. Rs. 153,680 c. Rs. 129,600 d. Rs. 119,900

$$1,60,000 \times \frac{90}{100} \times \frac{90}{100} = ?$$

$$16 \times 81 = 1296$$

13 The present population of a country estimated to be 10 crores is expected to increase to 13.31 crores during the next 3 years the uniform rate of growth is?

10%

$$10 \times (M.F)^3 = 13.31$$

$$(M.F) = \sqrt[3]{\frac{1331}{1000}} = \frac{11}{10}$$

$$\frac{+1}{10} \times 100 = 10\% \uparrow$$

14

The population of different trees in a field increased by 10 % in first year, increased by 8 % in second year and decreased by 10 % in third year. If at present the number of trees is 26730, then find the number of tress in the beginning.

a. 30000 b. 25000 c. 27000 d. 27865

$$P \times \frac{110}{100} \times \frac{108}{100} \times \frac{90}{100} = 26730$$

14

The population of different trees in a field increased by 10 % in first year, increased by 8 % in second year and decreased by 10 % in third year. If at present the number of trees is 26730, then find the number of tress in the beginning.

a. 30000 b. 25000 c. 27000 d. 27865

$$P \times \frac{\cancel{110}^{10}}{100} \times \frac{\cancel{108}^{\cancel{12}\ 4}}{\cancel{100}_{25}} \times \frac{9\cancel{0}}{100} = \cancel{2}\cancel{6}_{8}\cancel{7}3\cancel{0}$$

$\cancel{297}$

$\cancel{27}$

$\cancel{3}$